UCLA
Department of Statistics

STAT 208: STATISTICAL LEARNING THEORY

Problem Set 1
Winter 2018

Issued: Thursday, January 11 **Due:** Thursday, January 25

Full credit: 90 points.

Problem 1.1 (50 points)

In this problem, we derive a general approach—called *iterative reweighed least-squares*—for obtaining the ERM solution with a smooth loss, and unconstrained linear function class. Consider minimizing a function $L : \mathbb{R}^d \to \mathbb{R}$, and assume that L is continuously differentiable up to second order. The Newton iteration for obtaining the minimizer of L is

$$\theta_{t+1} = \theta_t - [\nabla^2 L(\theta_t)]^{-1} \nabla L(\theta_t) \tag{1}$$

where $\nabla^2 L$ and ∇L are the Hessian and gradient of L respectively.

(a) Show that iteration (1) is equivalent to minimizing the second-order Taylor expansion of L around θ_t.

(b) Let $L(\theta) = \frac{1}{n}\sum_{i=1}^n \ell(\theta^T x_i, y_i)$ where $\ell : \mathbb{R} \times \mathcal{Y} \to \mathbb{R}$ is smooth in its first argument. Argue that

$$\nabla L(\theta) = \frac{1}{n}\sum_{i=1}^n \alpha_i(\theta) x_i, \quad \nabla L^2(\theta) = \frac{1}{n}\sum_{i=1}^n w_i(\theta) x_i x_i^T$$

for weights $\alpha_i(\theta)$ and $w_i(\theta)$ that you specify.

(c) Let $X \in \mathbb{R}^{n\times d}$ be the matrix whose i row x_i^T. Let $\alpha(\theta) = (\alpha_1(\theta), \dots, \alpha_n(\theta))$ and $W_\theta = \mathrm{diag}(w_1(\theta), \dots, w_n(\theta))$. Show that iteration (1) is equivalent to solving

$$\theta_{t+1} = \underset{\theta\in\mathbb{R}^d}{\mathrm{argmin}}\, \frac{1}{2}\|W_{\theta_t}^{1/2}(X\theta - z(\theta_t))\|^2 \tag{2}$$

for $z(\theta_t) \in \mathbb{R}^n$ which you specify. Thus Newton iteration in this context is equivalent to solving a reweighed least-squares problem at each iteration.

Hint: $\nabla L(\theta) = X^T \alpha(\theta)$ *and* $\nabla^2 L(\theta) = X^T W_\theta X$.

(d) Consider the case where $\ell(t, y) = -yt + \log(1 + e^t)$ is the logistic loss. Show that in this case, $w_i(\theta) = \sigma(\theta^T x_i)(1 - \sigma(\theta^T x_i))$ and find an expression for $z_i(\theta)$.

Solution 1.1

(b) $\alpha_i(\theta) = \ell'(\theta^T x_i, y_i)$ and $w_i(\theta) = \ell''(\theta^T x_i, y_i)$

(c) $z(\theta) = X\theta + W^{-1}\alpha(\theta)$.

(d)

$$z_i(\theta) = x_i^T\theta + \frac{\sigma(\theta^T x_i) - y_i}{\sigma(\theta^T x_i)(1 - \sigma(\theta^T x_i))}$$

Problem 1.2

Consider logistic regression with linear function class $\{x \mapsto \theta^T x : \theta \in \mathbb{R}^d\}$.

(a) Show that maximum likelihood estimation of θ corresponds to minimizing $\theta \mapsto \frac{1}{n}\sum_{i=1}^n \ell(\theta^T x_i, y_i)$ with the logistic loss

$$\ell(t, y) = -yt + \log(1 + e^t), \quad t \in \mathbb{R}. \tag{3}$$

(b) Show that the logistic loss is convex in its first argument.

Problem 1.3

Consider the unconstrained risk minimization problem:

$$f^*(x) = \underset{f}{\operatorname{argmin}} \ \mathbb{E}[\ell(f(X), Y)].$$

(a) Show that the f^* is the minimizer of the posterior risk, i.e.,

$$f^*(x) = \underset{a \in \mathcal{A}}{\operatorname{argmin}} \ \mathbb{E}[\ell(a, Y) \mid X = x].$$

In each of the following cases, obtain the optimal Bayes rule. In each case the loss $\ell : \mathcal{A} \times \mathcal{Y} \to \mathbb{R}$ where the action space $\mathcal{A}$ could be different than the prediction space $\mathcal{Y}$.

(b) $\mathcal{Y} = \mathcal{A} = \mathbb{R}$ and ℓ the quadratic loss: $\ell(t, y) = (t - y)^2$.

(c) $\mathcal{Y} = \{0, 1\}$, $\mathcal{A} = \mathbb{R}$ and ℓ the logistic loss: $\ell(t, y) = -yt + \log(1 + e^t)$, $t \in \mathbb{R}$.

(d) $\mathcal{Y} = \{-1, 1\}$, $\mathcal{A} = \mathbb{R}$ and ℓ the hinge loss: $\ell(t, y) = (1 - yt)_+ = \max\{0, 1 - yt\}$, $t \in \mathbb{R}$.

(e) $\mathcal{Y} = \{-1, 1\}$, $\mathcal{A} = \mathbb{R}$ and ℓ the exponential loss: $\ell(t, y) = e^{-yt/2}$, $t \in \mathbb{R}$.

(f) $\mathcal{Y} = \mathcal{A} = \{0, 1\}$ and ℓ the asymmetric 0-1 loss:

$$\ell(a, y) = \begin{cases} \alpha & a = 1, y = 0 \\ 1 - \alpha & a = 0, y = 1 \\ 0 & \text{otherwise.} \end{cases}, \quad \alpha \in (0, 1).$$

Problem 1.4

Consider a K classification problem with class-conditionals given by multivariate Gaussian distributions $X \mid Y = k \sim N(\mu_k, \Sigma_k), k = 1, \dots, K$.

(a) Given the knowledge of μ_k and Σ_k, show that the optimal Bayes rule can be written as

$$f^*(x) = \operatorname*{argmax}_k \delta_k(x), \quad \delta_k(x) = -\frac{1}{2}\log|\Sigma_k| - \frac{1}{2}(x-\mu_k)^T\Sigma_k(x-\mu_k) + \log\pi_k.$$

This is called quadratic discriminant analysis. (The functions δ_k are called a discriminant functions.)

(b) Assuming $\Sigma_k = \Sigma$ for all $k \in [K]$, show that the optimal Bayes rule can be written as

$$f^*(x) = \operatorname*{argmax}_k \delta_k(x), \quad \delta_k(x) = x^T\Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^T\Sigma^{-1}\mu_k + \log\pi_k.$$

This is called linear discriminant analysis (LDA).

(c) Consider binary classification $\mathcal{Y} = \{0,1\}$, and show that the LDA of part (b) can be written as a thresholds linear/affine function, i.e., $f^*(x) = 1\{w^Tx + b > 0\}$

Problem 1.5

This problem investigate Fisher LDA, which can be considered a dimension reduction technique geared towards classification. The idea is to reduce the feature X to a single dimension by forming the projection $X_w := w^TX$, and then classify based on X_w. For example, in binary classification Fisher LDA uses the following rule $f_w(x) = 1\{w^Tx \geq \tau\}$ for some threshold τ.

The direction w is chosen so as to maximize the ratio of between-cluster variation to (average) with-cluster variation:

$$w^* = \operatorname*{argmax}_w \frac{\operatorname{var}(\mathbb{E}[X_w \mid Y])}{\mathbb{E}\operatorname{var}(X_w \mid Y)} \tag{4}$$

where $Y \in \{1,\dots,K\}$ is the categorical cluster label. As usual $(X,Y) \sim \mathbb{P}$. Assume that μ_k and Σ_k are the mean and covariance of X given $Y = k$, respectively, and let $\pi_k = \mathbb{P}(Y_k = k)$.

(a) Show that

$$w^* = \operatorname*{argmax}_w \frac{w^T\Sigma_B w}{w^T\Sigma_W w}$$

for matrices Σ_B and Σ_W that you specify. (B and W stand for "between" and "within".) This is a generalized eigenvalue problem in (Σ_B, Σ_W).

(b) Consider the binary classification $K = 2$. Show that $w^* = \pm\,\Sigma_w^{-1}(\mu_1 - \mu_2)$ up to scalar scaling.

(c) By replacing the population distribution $\mathbb{P}$ with the empirical distribution $\mathbb{P}_n = \frac{1}{n}\sum_{i=1}^n \delta_{(X_i,Y_i)}$ derive the sample version of Σ_B and Σ_W.

(d) Show that (4) is equivalent to

$$w^* = \operatorname*{argmax}_w \frac{\mathrm{var}(\mathbb{E}[X_w \mid Y])}{\mathrm{var}(X_w)}$$

Hint: Recall the law of total variance.

Solution 1.2

$\Sigma_B = \sum_k \pi_k(\mu_k - \bar{\mu})(\mu_k - \bar{\mu})^T$ where $\bar{\mu} = \sum_k \pi_k \mu_k$, and $\Sigma_W = \sum_k \pi_k \Sigma_k$.